

Hello.

How are you?

For Week 4 of Phase 2, I need to confirm the dataset-building and retraining rules.

1. For pseudo-labeling, can we finalize the automatic acceptance rule as:
confidence $\geq 95\%$
& no OCR conflict
& valid bounding box geometry
& no existing human-reviewed label for the same object/image?
2. Should we send a small percentage of accepted pseudo-labels to human review for quality control?
Example:
 95% accepted automatically
 5% randomly sampled for human QC
3. What lower training weight should pseudo-labeled samples use compared with human-reviewed labels?
Suggested options:
 Human-reviewed labels: 1.0
 Pseudo-labels: 0.2, 0.3, or 0.5
4. Can we use a separate directory structure for human labels, pseudo-labels, review batches, and merged training datasets?
5. What annotation format does the current training code expect?
6. For retraining, should the dataset contain only full Petri dish images, or should we also store cropped colony images?
7. Where should DVC store versioned datasets and artifacts?

8. How should active learning rounds and dataset versions be named?
9. Which validation metrics must be reported before a retrained model can be promoted?
10. What rule should decide whether a newly trained model is better than the current production model?
11. After builds the retraining dataset, should model retraining be triggered automatically, or should it require manual approval?
12. Should pseudo-labels be included in the same training run as human-reviewed labels, using lower loss weight?
13. If a pseudo-label conflicts with an existing human label or OCR result, what should happen?
14. For Week 4, what output format do you prefer for the final dataset builder deliverables?

Thank you.